CSE30301: Basic Math for AI - Final Exam

Student ID : **Name :**

Question 1 [30 points]

For each statement, mark **True (T)** or **False (F)**. You will receive +3 points for each correct answer, -2 points for each incorrect answer, and 0 points for unanswered questions.

1.1 For any differentiable scalar function $L(\theta)$, the gradient $\nabla L(\theta)$ points in the direction of steepest local increase. (T / F)

1.2 The negative gradient $-\nabla L(\theta)$ is always the fastest global path to the optimum. (T / F)

1.3 During gradient descent, if the update reaches a point where $\nabla L(\theta) = 0$, then the point must be a local minimum. (T / F)

1.4 In logistic regression with cross entropy loss, the gradient can be written as an error term multiplied by the input. (T / F)

1.5 In gradient descent, choosing a larger learning rate always decreases the number of iterations required for convergence. (T / F)

1.6 A small gradient norm always means that the model has already learned a good solution. (T / F)

1.7 Maximum likelihood estimation can be interpreted as minimizing the forward KL divergence from the empirical data distribution to the model distribution. (T / F)

1.8 KL divergence is symmetric, i.e., $D_{KL}(P||Q) = D_{KL}(Q||P)$ for all distributions P and Q. (T / F)

1.9 Reverse KL $D_{KL}(Q||P)$ is generally associated with mode-covering rather than mode-seeking. (T / F)

1.10 Wasserstein distance can still provide meaningful distance information when two distributions have disjoint supports. (T / F)

Question 2 [25 points]

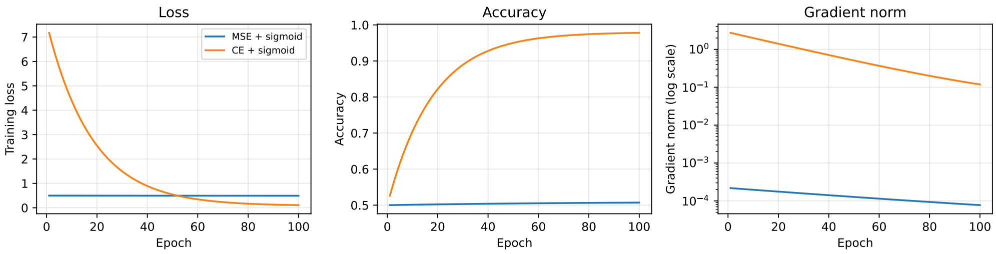
Consider binary classification with $x_i \in \mathbb{R}^d$, $y_i \in \{0, 1\}$, and

$$p_i = \sigma(x_i^T \beta), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

Two possible losses are

$$J_{MSE}(\beta) = \frac{1}{N} \sum_{i=1}^{N} (p_i - y_i)^2, \qquad J_{CE}(\beta) = -\frac{1}{N} \sum_{i=1}^{N} \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right].$$

The figure below summarizes one training behavior observed from the same bad initialization.



Training behavior of logistic regression with MSE loss and cross entropy loss.

2.1 Derive $\nabla_\beta J_{MSE}(\beta)$. (7 points)

2.2 Derive $\nabla_\beta J_{CE}(\beta)$. (8 points)

2.3 Using your results from **2.1** and **2.2**, explain why MSE can suffer from a vanishing-gradient problem under saturated wrong predictions, while cross entropy can still produce a useful gradient. (10 points)

Question 3 [20 points]

Consider the loss function $L(\theta_1, \theta_2) = \frac{1}{2}\theta_1^2 + \frac{1}{2}\theta_2^2$, and the current parameter $\theta = [3\ 4]^T$

Let $u = \begin{bmatrix} \cos\phi \\ \sin\phi \end{bmatrix}$ be a unit direction vector.

3.1 Compute $\nabla L(\theta)$ at $\theta = [3\ 4]^T$. (3 points)

3.2 Compute the directional derivative $D_u L(\theta) = \nabla L(\theta)^T u$ as a function of ϕ. (6 points)

3.3 Find the unit direction u^* that gives the fastest local decrease of the loss. (6 points)

3.4 Using your result above, explain why gradient descent uses the update: (5 points)

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t).$$

Question 4 [20 points]

Consider the one-dimensional quadratic objective $L(\theta) = a\theta^2$, where $a > 0$. We apply gradient descent:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \frac{dL}{d\theta}(\theta^{(t)}).$$

4.1 Find the minimizer θ^* of $L(\theta)$. Then compute $\frac{dL}{d\theta}$ and show that the gradient descent update can be written as $\theta^{(t+1)} - \theta^* = c(\theta^{(t)} - \theta^*)$. Write c in terms of a and η. (5 points)

4.2 Using your result from 4.1, express $\theta^{(t)} - \theta^*$ in terms of $\theta^{(0)} - \theta^*$ and c. (4 points)

4.3 For the sequence $\theta^{(t)}$ to converge to the minimizer θ^*, what condition should c satisfy? Use this condition to derive the range of η for convergence. (6 points)

4.4 Now consider the two-dimensional objective $L(\theta_1, \theta_2) = 10\theta_1^2 + \theta_2^2$. Based on the one-dimensional error dynamics above, explain why using a single learning rate can be difficult when different coordinates have different curvatures. (5 points)

Question 5 [30 points]

Consider the ill-conditioned quadratic objective

$$L(\boldsymbol{\theta}) = L(\theta_1, \theta_2) = 10\theta_1^2 + \theta_2^2, \text{ where } \boldsymbol{\theta}^{(t)} = \begin{bmatrix} \theta_1^{(t)} \\ \theta_2^{(t)} \end{bmatrix}$$

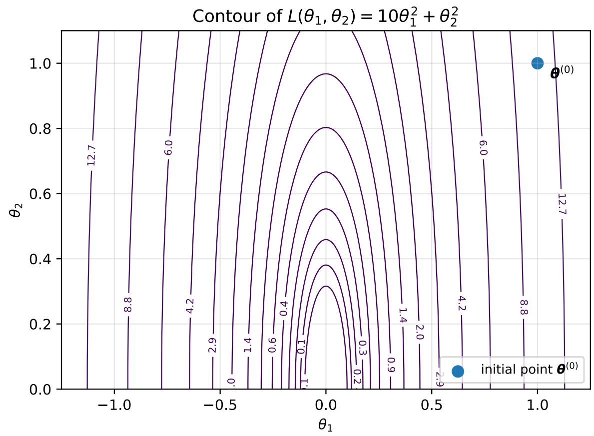
denotes the parameter vector at iteration t.

The vanilla gradient descent update is $\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \eta \nabla L(\boldsymbol{\theta}^{(t)})$.

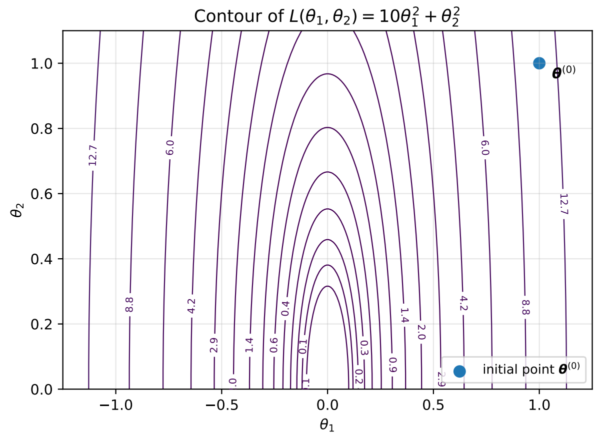
Suppose

$$\boldsymbol{\theta}^{(0)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \eta = 0.1.$$

The figure below shows the contour plot of $L(\theta_1, \theta_2)$ and the initial point $\boldsymbol{\theta}^{(0)}$. On the figure, you should draw the requested update trajectories by marking points and arrows.



Draw your solution for **5.1-5.2** using vanilla gradient update.



Draw your solution for **5.5** using the modified gradient update.

5.1 Compute $\nabla L(\boldsymbol{\theta}^{(0)})$ and the first gradient descent update $\boldsymbol{\theta}^{(1)}$. Then draw $\boldsymbol{\theta}^{(1)}$ and the arrow from $\boldsymbol{\theta}^{(0)}$ to $\boldsymbol{\theta}^{(1)}$ on the **left** figure. (3 points)

5.2 Compute $\boldsymbol{\theta}^{(2)}$ and $\boldsymbol{\theta}^{(3)}$ using the same gradient descent update. Draw $\boldsymbol{\theta}^{(2)}$, $\boldsymbol{\theta}^{(3)}$, and the corresponding arrows on the **left** figure. (5 points)

5.3 Diagnose and explain the main problem of this vanilla gradient update. (7 points)

5.4 To address this problem, propose a modified gradient update rule while keeping the same learning rate $\eta = 0.1$. Your update should use coordinate-wise scaling so that the effective update coefficients of the two coordinates become balanced. Specifically, design your update so that

$$\theta_1^{(t+1)} = 0.5\theta_1^{(t)}, \qquad \theta_2^{(t+1)} = 0.5\theta_2^{(t)}.$$

Clearly write your proposed update rule and explain why it reduces the problem observed in vanilla gradient descent. (8 points)

5.5 Starting from the same $\boldsymbol{\theta}^{(0)}$, compute the modified update results $\boldsymbol{\theta}^{(1)}$, $\boldsymbol{\theta}^{(2)}$, and $\boldsymbol{\theta}^{(3)}$. Draw these points and arrows on the **right** figure. (7 points)

Question 6 [20 points]

Let X be a continuous random variable with probability density function $p(x)$ satisfying

$$\mathbb{E}_p[X] = \mu, \qquad \mathrm{Var}_p(X) = \sigma^2.$$

For a continuous random variable, the entropy is defined as:

$$H(p) = -\int_{-\infty}^{\infty} p(x) \log p(x)\, dx.$$

Prove that, among all valid probability density functions with mean μ and variance σ^2, the Gaussian distribution $p^*(x) = \mathcal{N}(x; \mu, \sigma^2)$ achieves the maximum entropy.

Hint. Use the non-negativity of the KL divergence. The probability density function of Gaussian distribution $q(x) = \mathcal{N}(x; \mu, \sigma^2)$ is

$$q(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Question 7 [30 points]

Consider the following target distribution $p(x)$:

$$p(x) = \frac{1}{4}\mathcal{N}(x; -2, 1) + \frac{3}{4}\mathcal{N}(x; 2, 1), \quad \mathcal{N}(x; \mu, \sigma^2) := \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

We approximate $p(x)$ using a single Gaussian model distribution $q_{\mu,\sigma}(x) = \mathcal{N}(x; \mu, \sigma^2)$ by minimizing the forward KL divergence:

$$\min_{\mu,\sigma} D_{KL}(p||q_{\mu,\sigma}).$$

7.1 Show that the optimal mean satisfies $\mu^* = \mathbb{E}_p[X]$. (7 points)

7.2 Show that the optimal variance satisfies $(\sigma^*)^2 = \mathrm{Var}_p(X)$. (7 points)

7.3 Compute μ^*. (6 points)

7.4 Compute $(\sigma^*)^2$. (**Hint.** $\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X|Z)] + \mathrm{Var}(\mathbb{E}[X|Z])$) (10 points)

Question 8 [25 points]

Let the target distribution be $P(x) = \text{Unif}[0,1]$, and let the model distribution be $Q_\mu(x) = \text{Unif}[\mu, \mu+1]$. We consider the forward KL divergence

$$D_{KL}(P||Q_\mu) = \int P(x) \log \frac{P(x)}{Q_\mu(x)} dx,$$

and the Jensen-Shannon divergence

$$JSD(P, Q_\mu) = \frac{1}{2} D_{KL}(P||M) + \frac{1}{2} D_{KL}(Q_\mu||M), \qquad M = \frac{1}{2}(P + Q_\mu).$$

For this problem, let

$$a(\mu) = |[0,1] \cap [\mu, \mu+1]|$$

denote the overlap length between the supports of P and Q_μ.

8.1 For $\mu = \frac{1}{2}$, compute $D_{KL}(P||Q_\mu)$ and $JSD(P, Q_\mu)$. (10 points)

8.2 In the case $\mu = \frac{1}{2}$, which similarity measure, KL or JSD, is more appropriate for optimization and why? If neither is appropriate, explain precisely why each one is problematic. (7 points)

8.3 Now suppose the two supports are fully separated, for example $\mu = 2$. Which similarity measure, KL or JSD, is more appropriate for optimization and why? If neither is appropriate, explain precisely why each one is problematic. (8 points)